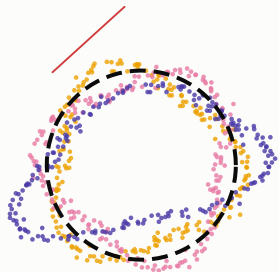


a

Fixed coordinate system

residual read as noise

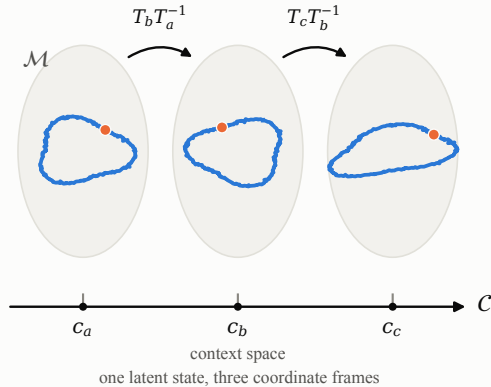


● ctx a ● ctx b ● ctx c

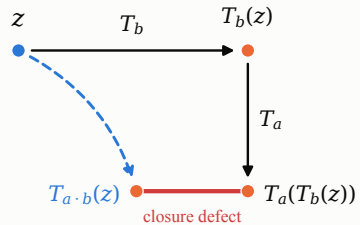
one manifold fitted to pooled activity

b

Gauge neural dynamics

**c**

Approximate group structure



$$T_{a \cdot b} = \exp \text{BCH}(\theta_a, \theta_b),$$

$$[G_i, G_j] = \sum_k f_{ij}^k G_k + R_{ij}$$